

RIGEL · SPEC EDITOR

Game Modes Guide

Build team games: teams, rounds, scores, traps, balls, scoreboards and your own Luau.

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1 What a game mode is

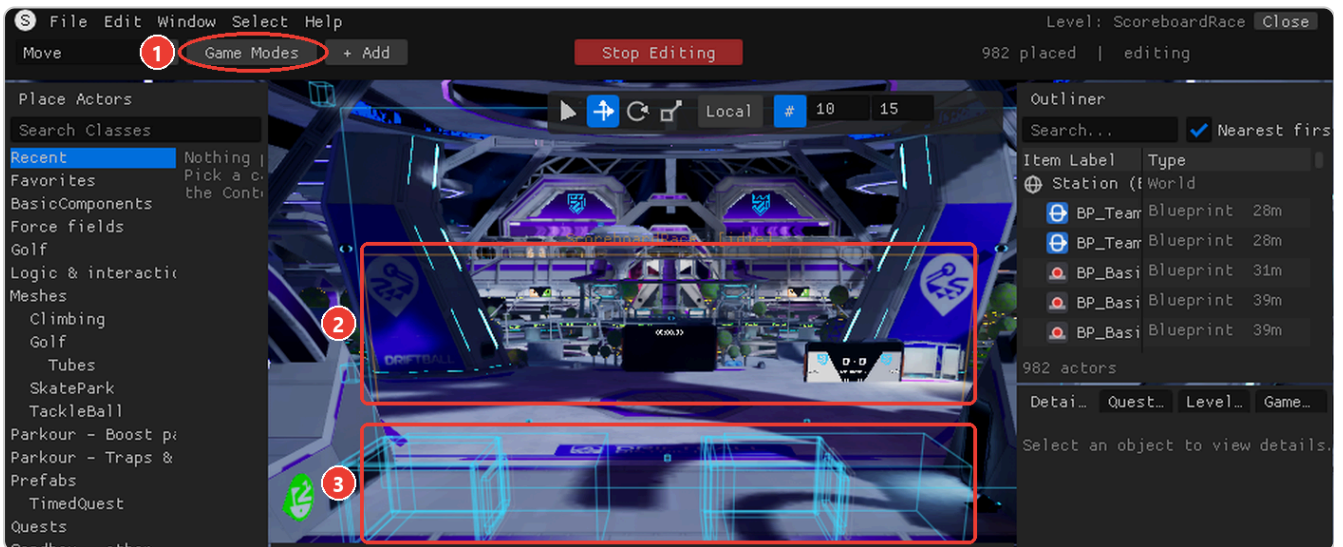
A **game mode** is an area of the station with its own **teams, rounds and scores**. You build one in the Spec Editor, fill it with pieces -- team changers, start and score buttons, traps, force fields, timers, ball spawners, scoreboards, text signs -- and the server runs it for everyone, Quest players included, with no mods on their side.

- Players pick a team by walking into one of the mode's **team changers**.
- A round goes **idle** -> **countdown** -> **running** -> **ended** -> **idle**. The server counts the countdown and the round time, keeps the scores, decides the winner and tells every machine.
- Pieces get **roles** ("start button", "trap on during rounds", "ball spawner" ...). The editor turns the roles into the mode's **controller script** and wires everything for you.
- Your own Luau adds the rules only you can think of, through a handful of hooks and the **Rigel** library.
- Game modes are saved in your levels and come back exactly when a level loads.

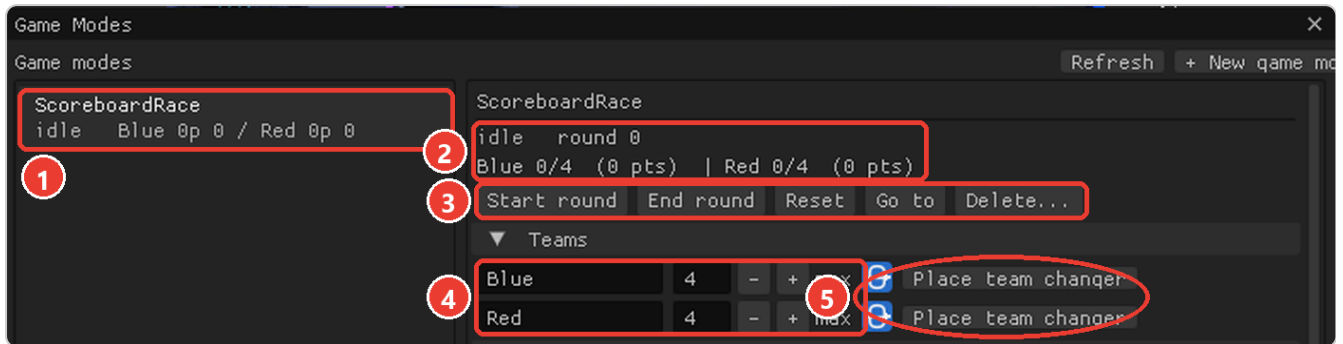
2 Opening the Game Modes window

Click **Game Modes** in the toolbar (next to **+ Add**), or open **Details > Game Modes** and click **Open the Game Modes window**. The window can be moved and resized; your game modes are on the left, the one you picked on the right. Every game mode also shows in the viewport as an orange box (green while a round runs) with its name and state.

A level must be open to make a game mode (**Levels** tab, or Ctrl+S to save one).



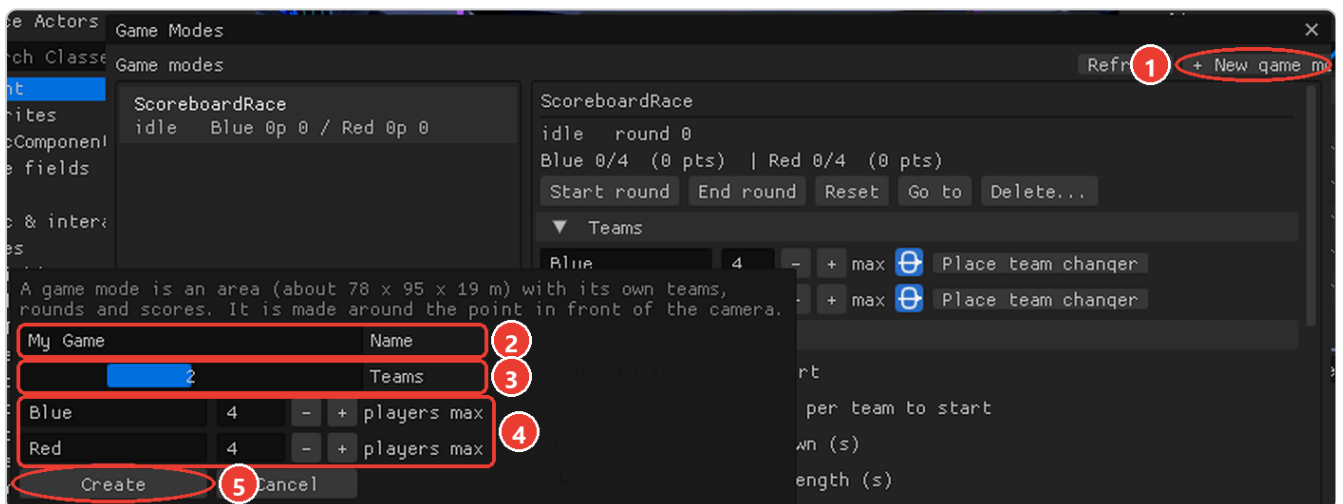
1 the **Game Modes** button in the toolbar. **2** a game mode's area: the orange box, with its name and state on top. **3** its team changers (outlined).



The Game Modes window. **1** your game modes (click one to edit it). **2** its state, round, and each team's players and points. **3** round controls: Start round, End round, Reset, Go to (flies the camera there), Delete. **4** each team's name and max players. **5** Place team changer, one per team.

3 Making one

1. Fly to where the arena should be: a game mode is made around the point in front of the camera.
2. Click **+ New game mode**. Type a **name**, choose **1 to 4 teams**, name each team and set its **max players**.
3. Click **Create**. The orange box shows the area (about 78 x 95 x 19 m). Everything placed inside it belongs to the mode.



1 + New game mode. **2** the mode's name. **3** how many teams. **4** each team's name and max players. **5** Create.

Two modes can't overlap. **Delete...** removes a mode and everything in it.

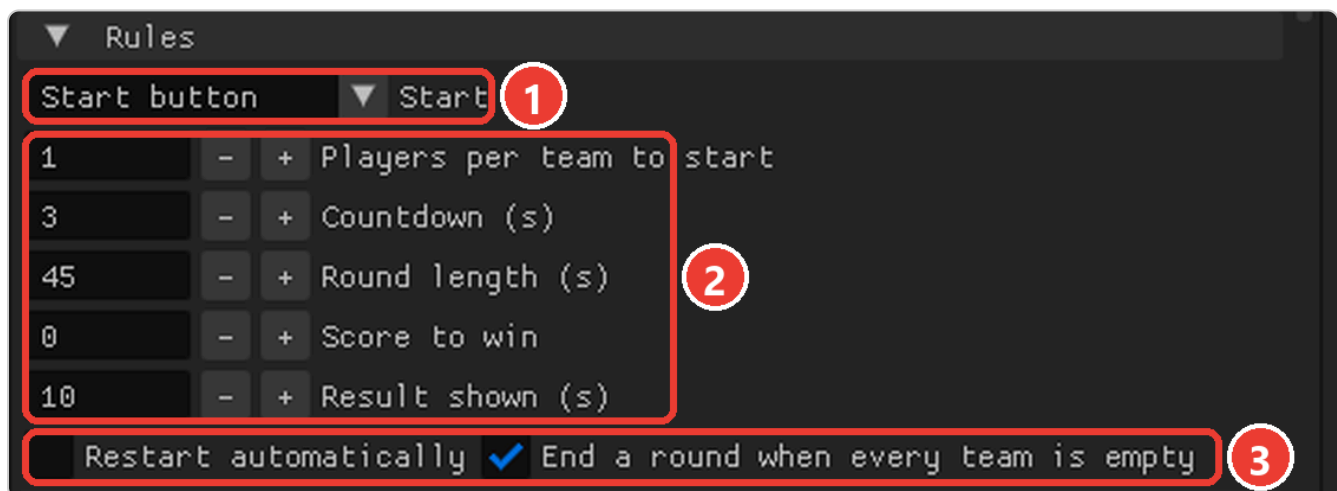
4 Teams and team changers

For each team, **Place team changer** puts a team changer in front of the camera (the spot must be inside the box). Walking into it puts a player on that team; the team's size and the max show in the window and on scoreboards.

- Rename a team or change its max right in the window (press Enter / click away to apply).
- Team changers only work inside a game mode. Placed anywhere else the server refuses them.
- If a player is standing exactly where you place (or move) a team changer, it appears next to them and moves into place as soon as they step off. A changer created on top of a player would crash their game.

5 Rules

Rule	Meaning
Start	<i>Manual</i> (the window's Start round, or a script), <i>Start button</i> (a button with the Start role), <i>Automatic</i> (when every team has enough players)
Players per team to start	for Automatic
Countdown (s)	seconds between "start" and the round beginning; 0 = straight away
Round length (s)	0 = no time limit (the round ends by score or by a script)
Score to win	a team reaching it wins at once; 0 = no limit
Result shown (s)	how long the result stays before the lobby
Restart automatically	start the next round after the result
End a round when every team is empty	so an abandoned round doesn't run forever: once a round has had players on a team and they all leave. A round started with empty teams keeps running
Ball back to its spawner after a goal	a Goal / Score box point resets every ball (a kick-off)



The rules. **1** how a round starts (Manual / Start button / Automatic). **2** players per team to start, countdown, round length, score to win, how long the result shows. **3** restart automatically, and end a round when every team is empty.

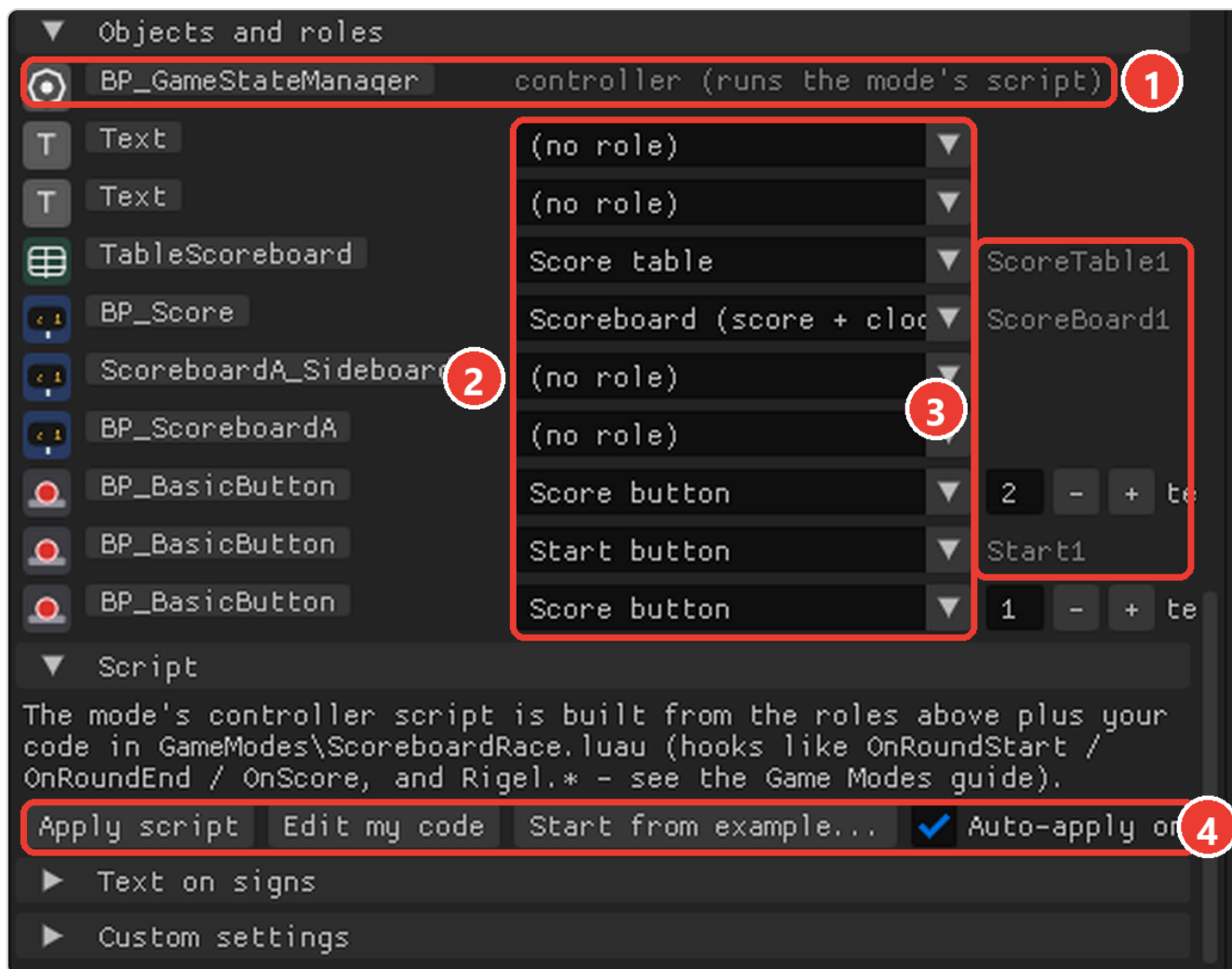
Start round, **End round** and **Reset** in the window control the mode by hand. **Go to** flies the camera there.

6 Pieces and roles

Place pieces shows every useful piece with its icon; clicking one places it inside the mode (in front of the camera) and gives it its role straight away. **Objects and roles** lists everything in the mode with its icon; change any object's role from its drop-down.



Place pieces: each tile places that piece inside the mode with its role already set. **1** buttons: start, score, trap. **2** traps (the whole orange group). **3** moving platforms. **4** force field, for walls that open and close. **5** the round timer. **6** timer display, scoreboards and the score table.



Objects and roles. **1** the controller: the object that runs the mode's script (made with the mode, keep it). **2** each object's role. **3** the slot name your own code uses for that object (Start1, ScoreBoard1 ...). Score buttons also say which team they score for. **4** Apply script, Edit my code, Start from example..., and Auto-apply on save.

Role	For	What it does
Start button	buttons	starts a round (when none is on)
Score button (team N)	buttons	+1 point for team N while a round runs
Trap - on during rounds	traps, sliding platforms	switched on when a round starts, off when it ends
Trap - pulses during rounds (seconds)	traps	flips on/off every N seconds while a round runs
Trap - fired by trap buttons	traps	switched on for a while when a trap button is pressed
Trap button (seconds)	buttons	fires every "fired by trap buttons" trap for N seconds
Wall - between rounds only	force fields, walls	solid between rounds, gone while a round runs
Wall - during rounds only	force fields, walls	only there while a round runs

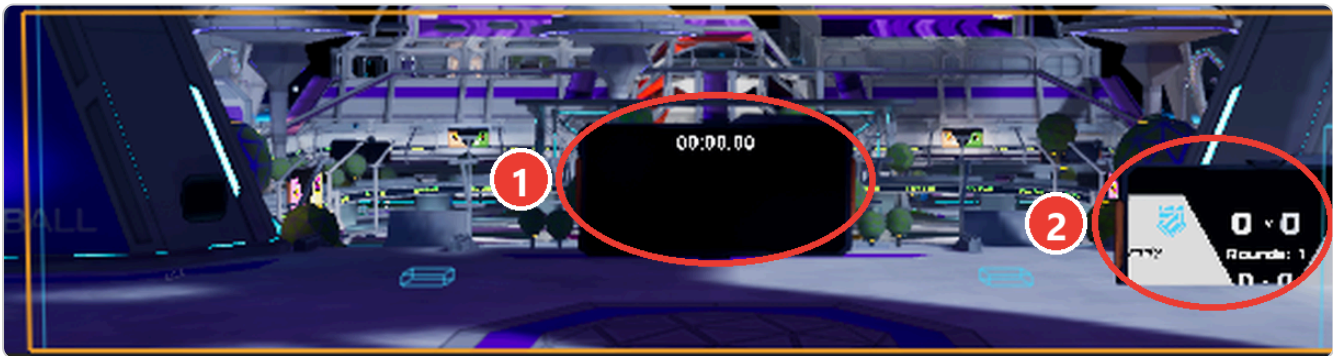
Force fields are switched on and off for the wall roles; any other piece is shown and hidden (and its collision switched) instead -- the editor picks the right one for the object.

Round timer	Timer	counts the countdown, then the round (or up, with no limit)
Ball spawner	Jake ball / ball spawners	the server spawns or resets its ball when a round starts
Goal (team N)	Driftball goal, Goal	the ball going in during a round gives team N the goal's points (1, more for long shots)
Score box (team N, 0 = by side)	Score box (a disc trigger)	the ball flies <i>through</i> it; each pass during a round gives team N a point. 0 = by side: along the box's forward arrow scores for team 1, the other way for team 2
Ball start ring	Ball start ring	between rounds, carry the ball into it to start a round; hidden while a round is on
Scoreboard (score + clock)	the classic Score board	shows team 1 / team 2 points
Score table	the Score table	one row per team: name, score, players, rounds won

After changing roles click **Apply script** (or keep **Auto-apply on save** on and just save your code).

Scoreboards

- **Scoreboard monitors** (and the side / half-court versions) need no role: the mode keeps its team scores and rounds won on them.
- The classic **Score board** shows team 1 and team 2 points (role *Scoreboard*).
- The **Score table** shows a row per team (role *Score table*).
- A **Text sign** can show anything the mode knows through templates (below).



In the arena: **1** a scoreboard monitor (its clock and the teams' scores), **2** the classic Score board (team 1 : team 2, rounds).

Balls, goals and the start ring (driftball)

Jake ball spawners (and the other ball spawners) spawn a real, networked ball on the server that every player sees. Give the spawner the **Ball spawner** role: the server spawns or resets the ball when each round starts, and `Rigel.resetBalls()` does it whenever your code asks (after a goal, say). Don't spawn balls from Luau: the script runs on every machine and each would make its own ball that only it can see.

- **Goals.** Place a *Driftball goal* from the pieces (or give any goal the **Goal** role) and set the team that scores there. The game's own goal detection fires, the mode adds the goal's points to that team, and -- with *Ball back to its spawner after a goal* on (the default) -- every ball goes back to its spawner for a kick-off. The goal takes the colour of the team defending it.
- **Score boxes.** A *Score box* is a trigger the ball passes through (it doesn't bounce off). Give it a team, or **0 = by side**: the direction the ball crosses decides the team (along the box's forward arrow = team 1, the other

way = team 2) -- rotate the box so its arrow points at team 2's end. Scale it to the gap you want to guard.

- **Start ring.** With a *Ball start ring* the mode keeps its ball out between rounds; a player carries it into the ring to start the round, like the station's arenas. The first 3 seconds after a round ends don't count (the ball is being put back), so a ring next to the spawner is fine. The ring hides while a round runs.
- A goal or box counts once even though every machine sees it (the server ignores repeats of the same goal within 2.5 s), and only while a round is running. In your own code, `Rigel.goal(team, points, key)` does the same.

7 Text on signs

Put a Text object in the mode and give it a text with `{templates}`. They update live for everyone. For a mode called **Arena**:

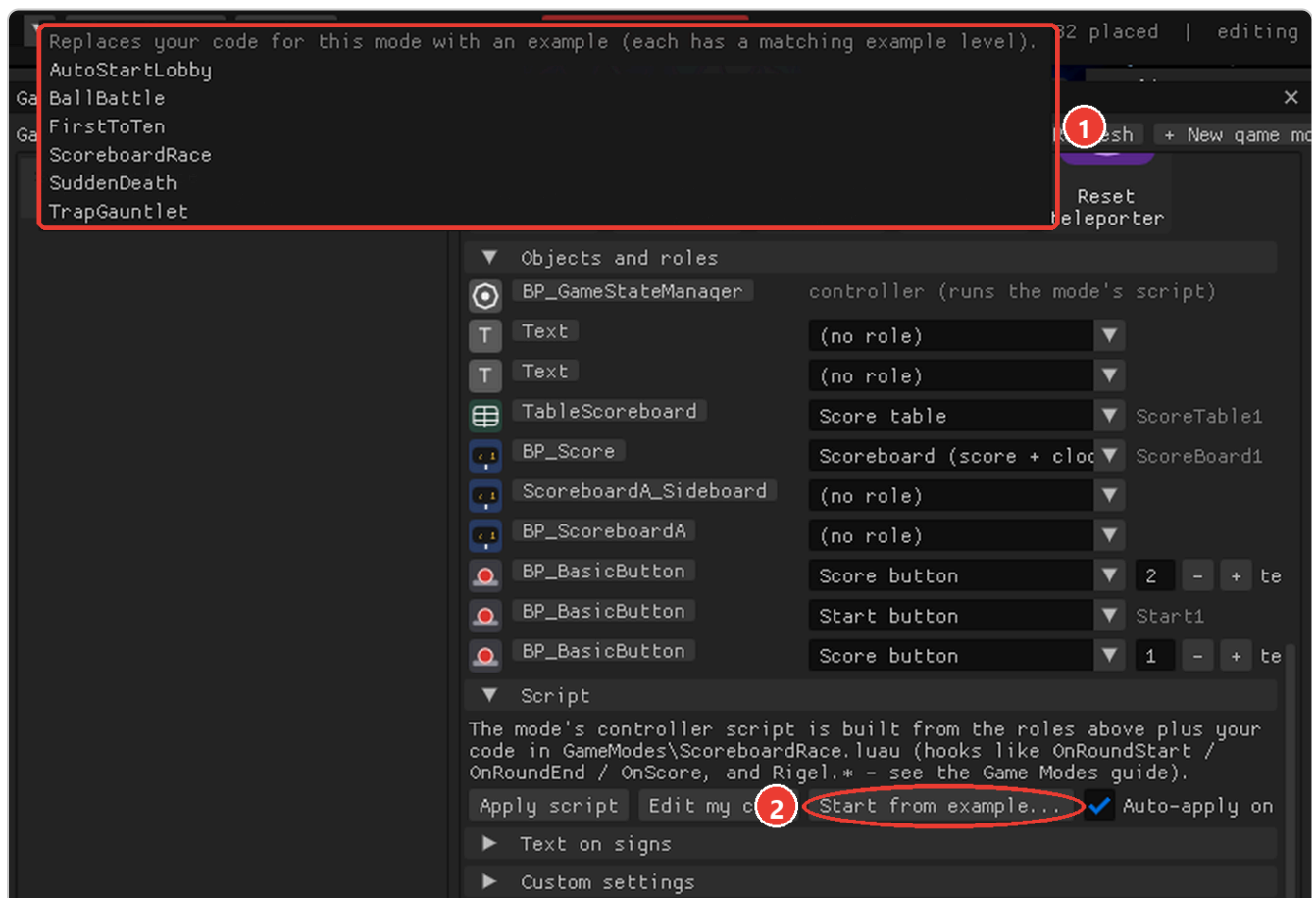
```
{Arena.state} {Arena.time} {Arena.round} {Arena.winner} {Arena.players} {Arena.team1.name}  
{Arena.team1.score} {Arena.team1.size} {Arena.team1.max} {Arena.team1.wins} (and team2, ...)
```

The window's **Text on signs** section lists them with a copy button. Your code can add its own:

```
Rigel.setText("Leader", "Blue lead!") fills {Leader}.
```

8 Your own code

Click **Edit my code** to open `Documents\RigelScripts\GameModes\<mode name>.luau` in VS Code (a template is made the first time), or **Start from example...** to begin from one of the examples. The editor puts your code into the mode's controller script together with the roles; **Show generated** shows the whole thing.



2 Start from example... opens 1 the list of example codes. Picking one replaces your code for this mode and applies it.

Don't write `BeginPlay` -- write any of these hooks:

```
function OnLobby() end -- no round on
function OnCountdown(seconds: number) end -- a round is about to start
function OnRoundStart(round: number) end -- round = 1, 2, ...
function OnRoundEnd(winner: number) end -- winner = team number, 0 = draw
function OnTeamChanged(team: number, size: number, old: number) end
function OnScore(team: number, score: number, old: number) end
function OnTime(secondsLeft: number) end -- every second of a countdown or round
```

Objects with roles are available by their slot names (shown next to each object in the window): `Start1`, `Score1`, `TrapRound1`, `TrapFired2`, `Timer1`, `Ball1`, `ScoreBoard1`, ...

The Rigel library

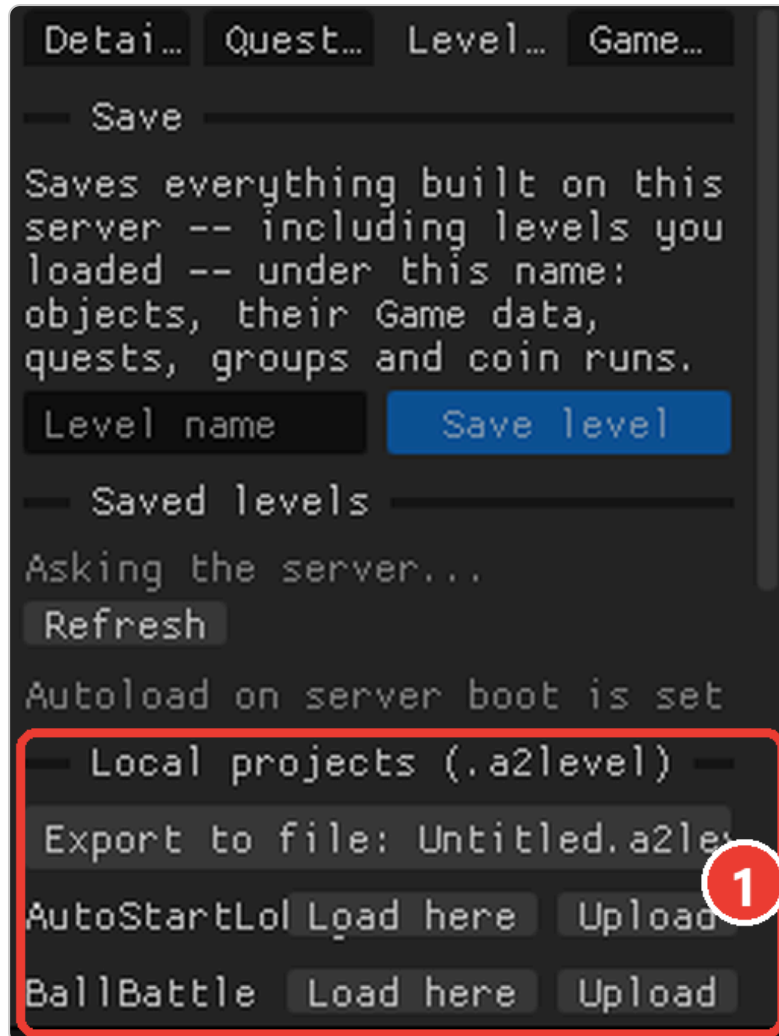
Read	
<code>Rigel.state()</code>	"idle", "countdown", "running", "ended"
<code>Rigel.isRunning()</code> , <code>Rigel.round()</code> , <code>Rigel.timeLeft()</code>	round state; seconds left
<code>Rigel.teams()</code> , <code>Rigel.teamName(t)</code> , <code>Rigel.teamSize(t)</code> , <code>Rigel.teamMax(t)</code>	teams are numbered from 1
<code>Rigel.score(t)</code> , <code>Rigel.roundsWon(t)</code> , <code>Rigel.players()</code> , <code>Rigel.winner()</code>	
<code>Rigel.setting("round_time")</code> , <code>Rigel.settingNumber(k)</code>	the rules, and your Custom settings

Do	
<code>Rigel.startRound()</code> , <code>Rigel.startNow()</code>	with / without the countdown
<code>Rigel.endRound(winner?)</code>	a team number, or nothing to decide by score
<code>Rigel.resetGame()</code>	scores and rounds back to zero
<code>Rigel.addScore(team, n?)</code> , <code>Rigel.setScore(team, n)</code>	points count only during a round
<code>Rigel.goal(team, n?, key?)</code>	a goal: counted once per <code>key</code> per 2.5 s, then balls reset (rule <i>Ball back to its spawner</i>)
<code>Rigel.setCustom(key, value)</code> , <code>Rigel.setText(name, value)</code> , <code>Rigel.resetBalls()</code>	

Scripts run on every machine; the Rigel actions only take effect on the server, so it's safe that every machine calls them. `Rigel.onStateChanged(f)`, `onTeamChanged(f)`, `onScoreChanged(f)`, `onTimeChanged(f)` are the callbacks behind the hooks -- use them from any other script in the mode too.

9 The examples

Each example comes as a **level** (Levels tab, *Local projects*) with the arena built -- team changers, buttons, traps, scoreboards, signs -- and as code (**Start from example...**):



Levels tab: **1** the example levels are under *Local projects*. Click **Load here** to open one on the server you're on.

Level	What it shows
FirstToTen	a score race with a live "who leads" sign, start + score buttons, a scoreboard monitor and a Score board
AutoStartLobby	a lobby that counts players in and starts by itself, a force field that opens for the round, a timer
SuddenDeath	a timed round; a draw goes into a 30-second sudden death; a Score table
TrapGauntlet	traps fired by a trap button, a pulsing laser, a spinner on during rounds, all fired for the last 20 s
BallBattle	a Jake ball match: first to 5, the ball resets after every point
ScoreboardRace	every kind of scoreboard and a catch-up rule for the last 15 seconds

10 Troubleshooting

- **"Team changers only work inside a game mode"**: place them inside the orange box.
- **The round ends by itself**: *End a round when every team is empty* is on and everyone who was on a team left.
- **A button does nothing**: check its role in *Objects and roles*, then **Apply script**. Score buttons only count during a round.

- **My code doesn't run:** the Problems popup shows script errors; the game log shows `[GM_<mode>.lua]` lines. Use the checker in VS Code (the kit's *Rigel checks*), and never write `BeginPlay` in mode code.
- **The ball doesn't come back:** give the spawner the *Ball spawner* role and call `Rigel.resetBalls()`.

11 Building game modes with an AI agent

The editor has an MCP server (`RigelScripts\mcp\rigel_mcp.py`) that lets an AI agent such as Claude or Codex build game modes, place objects and write, check and fix Luau for you. See **Rigel-MCP-Guide.pdf**.